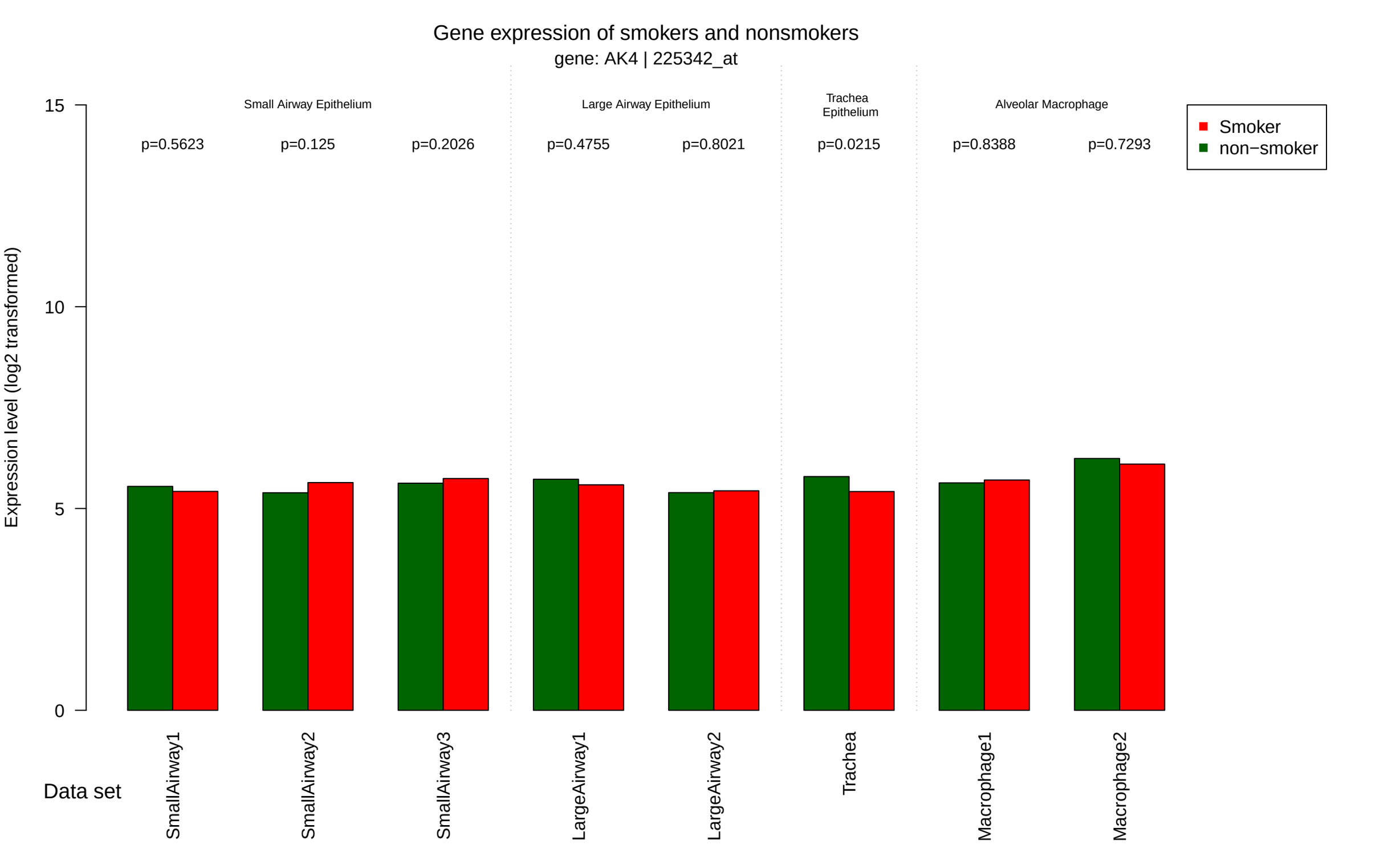
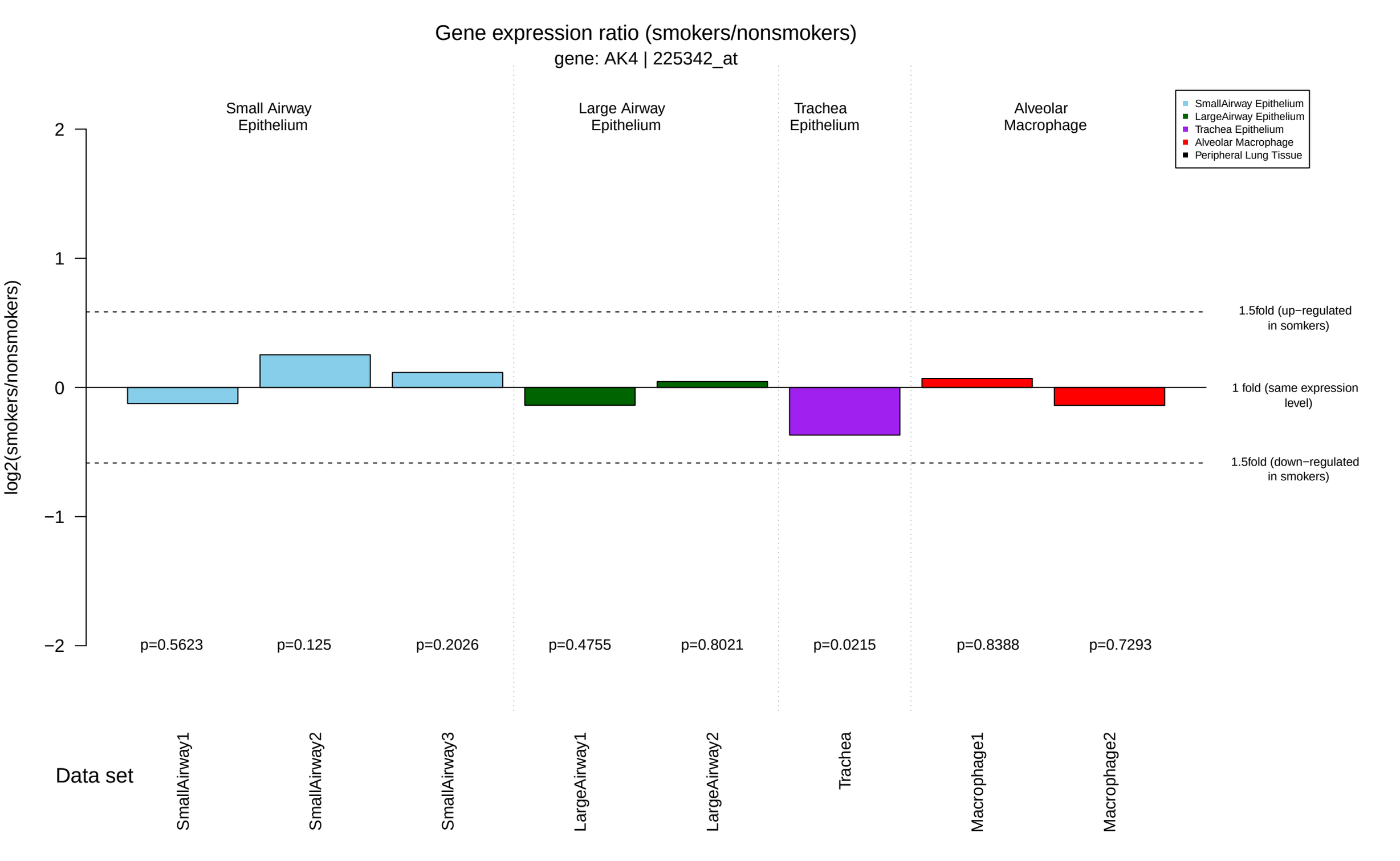
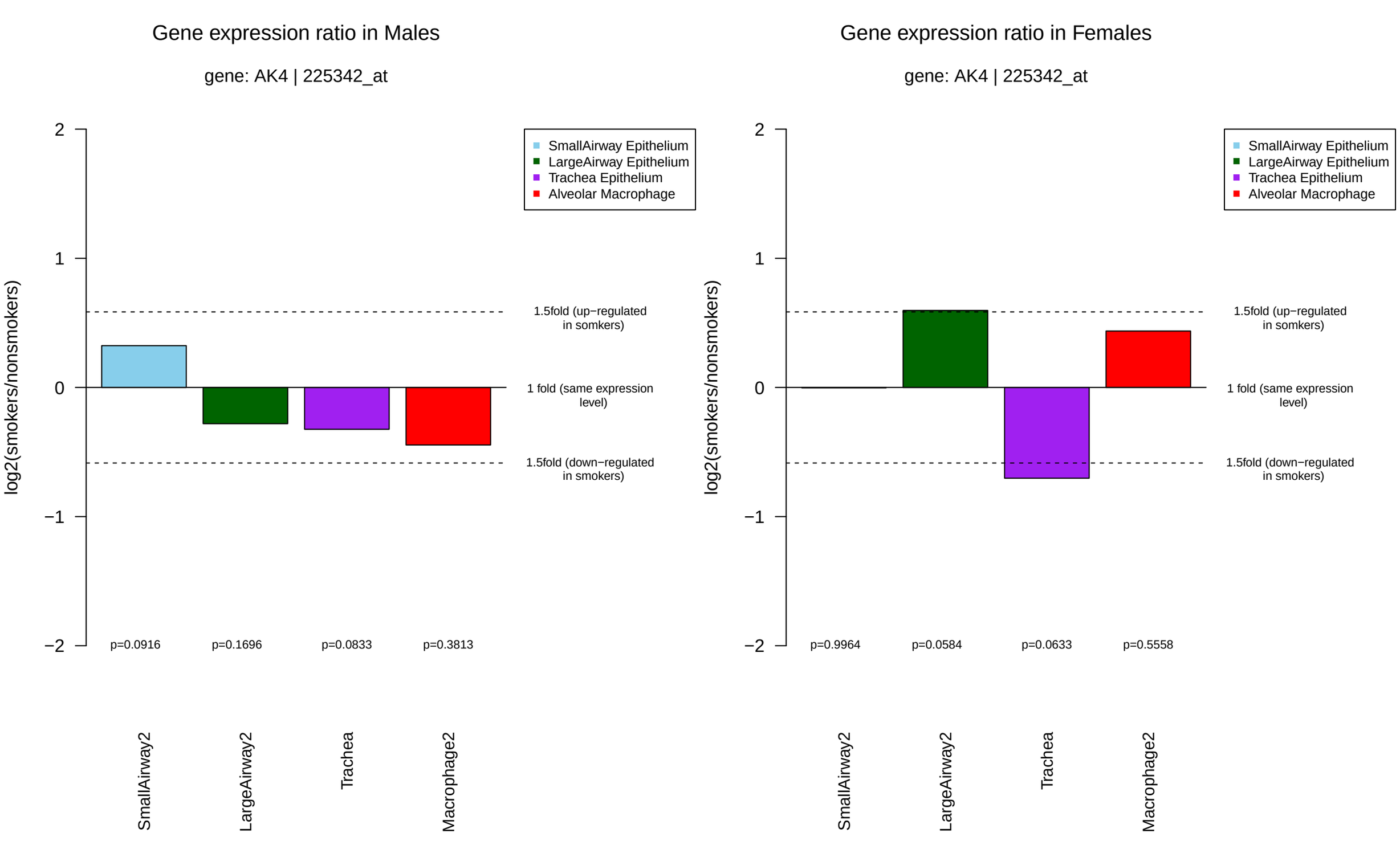
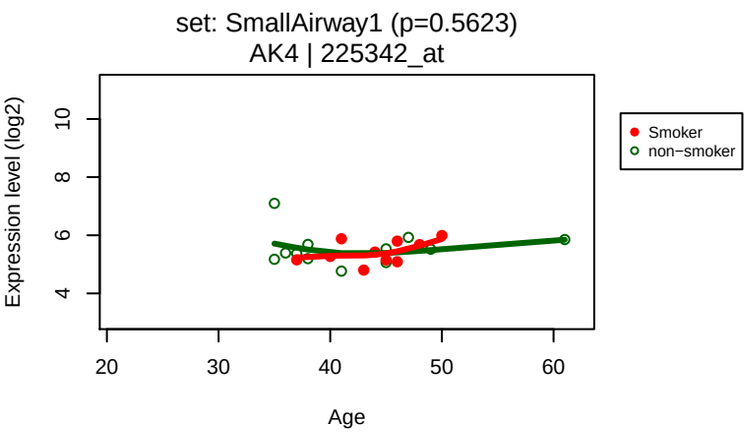
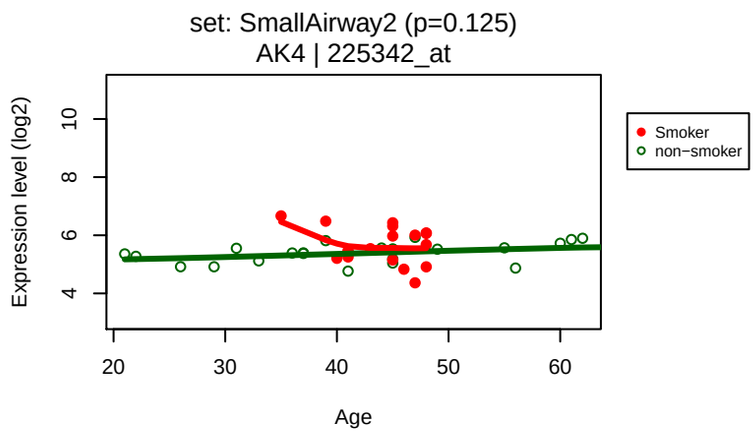
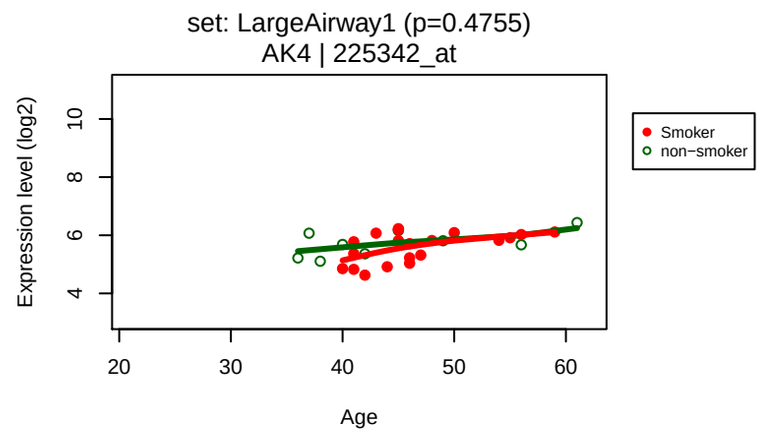

Gene expression of smokers and nonsmokers
gene: AK4 | 225342_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5623
p=0.125
p=0.2026
p=0.4755
p=0.8021
p=0.0215
p=0.8388
p=0.7293
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

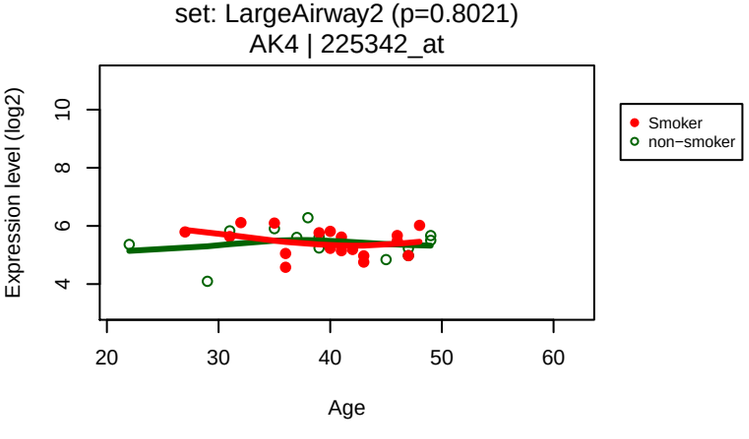

Gene expression ratio (smokers/nonsmokers)
gene: AK4 | 225342_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5623
p=0.125
p=0.2026
p=0.4755
p=0.8021
p=0.0215
p=0.8388
p=0.7293
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

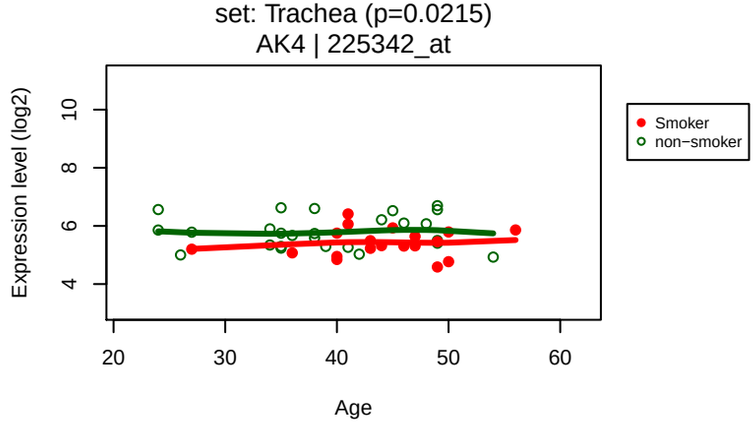

Gene expression ratio in Males
gene: AK4 | 225342_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0916
p=0.1696
p=0.0833
p=0.3813
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AK4 | 225342_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9964
p=0.0584
p=0.0633
p=0.5558
SmallAirway2
LargeAirway2
Trachea
Macrophage2

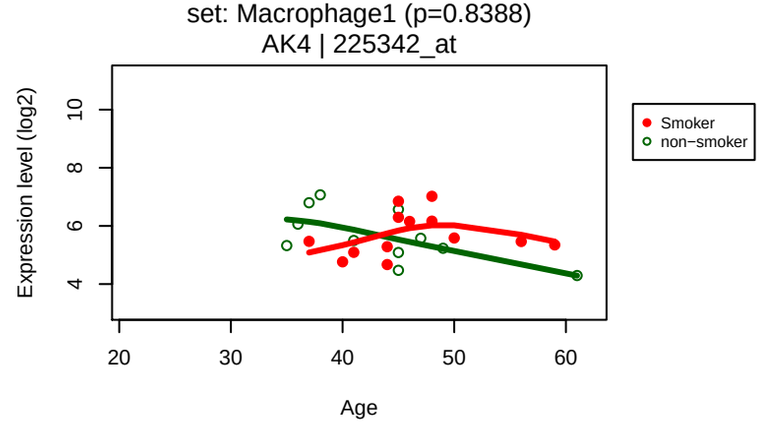

set: SmallAirway1 (p=0.5623)
AK4 | 225342_at
Expression level (log2)
Age
Smoker
non−smoker

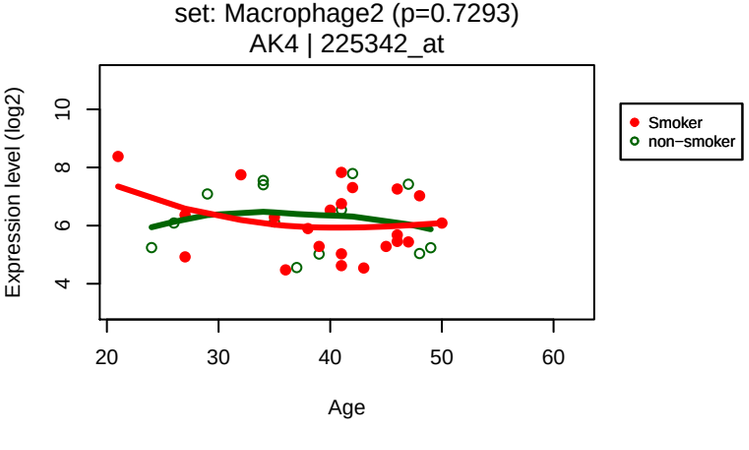

set: SmallAirway2 (p=0.125)
AK4 | 225342_at
Expression level (log2)
Age
Smoker
non−smoker

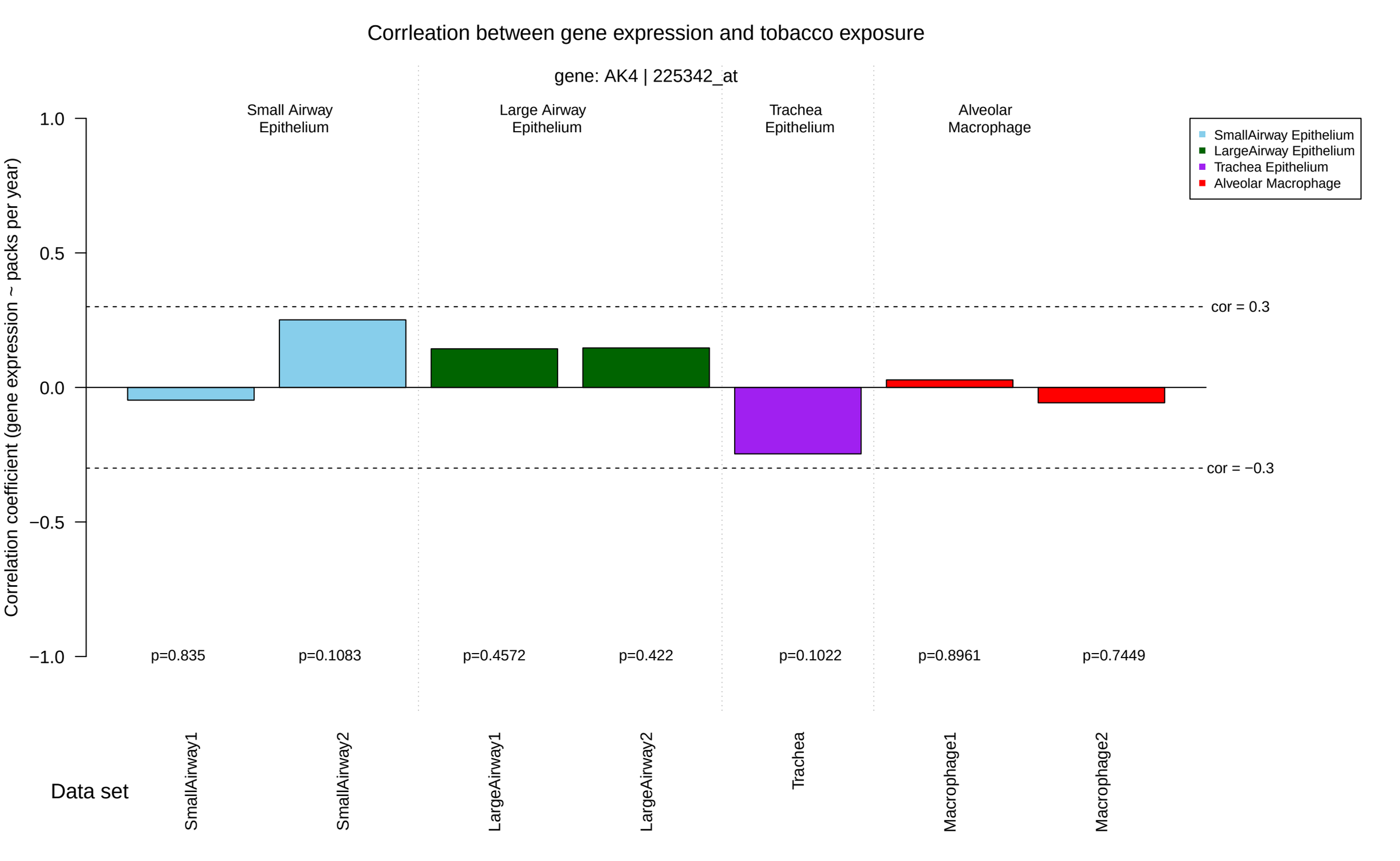

set: LargeAirway1 (p=0.4755)
AK4 | 225342_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8021)
AK4 | 225342_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0215)
AK4 | 225342_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8388)
AK4 | 225342_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7293)
AK4 | 225342_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AK4 | 225342_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.835
p=0.1083
p=0.4572
p=0.422
p=0.1022
p=0.8961
p=0.7449
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2